

SymPy Bug Report

1 Bug 15: Uniform Quantile Misses the $p = 1$ Endpoint

1.1 Minimal Reproducer

```
from sympy import FiniteSet, S
from sympy.stats import P, Uniform, cdf, quantile

U = Uniform("U", 0, 1)
actual = quantile(U)(S.One)
expected = FiniteSet(S.One)

print('quantile(Uniform("U", 0, 1))(1) =', actual)
print('cdf(Uniform("U", 0, 1))(1) =', cdf(U)(S.One))
print("P(U <= 1) =", P(U <= 1))
print("expected p=1 quantile =", expected)
print("matches expected =", actual == expected)
```

1.2 Observed Behavior

```
quantile(Uniform("U", 0, 1))(1) = EmptySet
cdf(Uniform("U", 0, 1))(1) = 1
P(U <= 1) = 1
expected p=1 quantile = {1}
matches expected = False
```

SymPy reports that the $p = 1$ quantile is the empty set, even though its own CDF and probability computations confirm that the uniform random variable reaches probability 1 at the upper endpoint.

1.3 Expected Behavior

For $U \sim \text{Uniform}(0, 1)$, the set-valued quantile API should return $\{1\}$:

$$\text{quantile}(U)(1) = \{1\}.$$

Equivalently, in SymPy's representation, the expected result is `FiniteSet(S.One)`.

1.4 Mathematical Explanation

SymPy documents the quantile as

$$Q(p) = \inf\{x \in \mathbb{R} : p \leq F(x)\},$$

where F is the CDF. For $U \sim \text{Uniform}(0, 1)$,

$$F(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

At $p = 1$, the set $\{x : 1 \leq F(x)\}$ is $[1, \infty)$, so its infimum is 1. The result $\emptyset$ is therefore not an equivalent representation, a harmless branch choice, or a missing simplification: it says that no endpoint value satisfies the quantile request, contradicting the documented infimum definition and the computed facts $F(1) = 1$ and $P(U \leq 1) = 1$.

1.5 Source-Level Diagnosis

The diagnosis identifies the relevant generic implementation in `sympy/stats/crv.py`, specifically the `compute_quantile` method around lines 285–291. The public `quantile` function delegates to the random variable’s probability space distribution. The class-level `quantile` method around lines 296–302 first tries a distribution-specific `_quantile`, but `UniformDistribution` does not provide one, so the call falls back to `compute_quantile`. The documented quantile definition is in `sympy/stats/rv.py` around lines 1198–1203.

The generic implementation computes a CDF expression and then solves the equality $F(x) = p$ over the distribution support:

```
cdf = integrate(pdf, (x, left_bound, x), **kwargs)
quantile = solveset(cdf - p, x, self.set)
return Lambda(p, Piecewise((quantile, (p >= 0) & (p <= 1)), (nan, True)))
```

For `Uniform(0, 1)`, the traced generic computation produced:

```
Lambda(_p, Piecewise((Intersection({_p}, Interval.Ropen(0, 1)),
                      (_p >= 0) & (_p <= 1)), (nan, True)))
```

This equality-solving path represents the inverse CDF as an intersection with the right-open interval $[0, 1)$ in the trace, written by SymPy as `Interval.Ropen(0, 1)`. Substituting $p = 1$ therefore drops the endpoint and evaluates to $\emptyset$. The source-level behavior causes the mathematical failure because the quantile is not merely the solution set of $F(x) = p$; at an endpoint it must follow the documented infimum definition. A plausible fix direction supported by the diagnosis is to implement the generic quantile through the inequality-and-infimum definition, or to add endpoint handling after solving so that $p = 1$ returns the support supremum when the CDF at that endpoint is 1.

1.6 Additional Failing Instances

The independent verification checks the same endpoint property for six finite uniform intervals. It computes the uniform CDF directly using rational arithmetic and verifies that a point just below the upper endpoint has CDF less than 1, while the endpoint itself has CDF exactly 1. Thus each case has the same expected behavior: the $p = 1$ quantile is the interval’s upper endpoint.

Input / Expression	SymPy behavior	Expected behavior
$\text{Uniform}(0, 1), p = 1$	$\emptyset$	$\{1\}$
$\text{Uniform}(-2, 3), p = 1$	$\emptyset$	$\{3\}$
$\text{Uniform}(2, 5), p = 1$	$\emptyset$	$\{5\}$
$\text{Uniform}(1/3, 7/3), p = 1$	$\emptyset$	$\{7/3\}$
$\text{Uniform}(-5/2, -1/2), p = 1$	$\emptyset$	$\{-1/2\}$
$\text{Uniform}(-1, 0), p = 1$	$\emptyset$	$\{0\}$

1.7 Related Issues Assessment

The best-effort deduplication check lists public Issue #21098 as related material and assigns the verdict `family_known_specific_new`. Based on the available evidence, this should be framed as a new instance of a known failure mode: the general endpoint fragility of quantile computation may be broadly understood, but this exact reproducible Uniform $p = 1$ endpoint case is previously unreported and remains individually actionable as a focused issue or regression test.

1.8 Proposed SymPy Regression Test

```
import pytest

from sympy import FiniteSet, Rational, S
from sympy.stats import Uniform, quantile

@pytest.mark.parametrize(
    "name,a,b",
    [
        ("unit_interval", S.Zero, S.One),
        ("shifted_interval", S(-2), S(3)),
        ("positive_interval", S(2), S(5)),
        ("rational_interval", Rational(1, 3), Rational(7, 3)),
        ("negative_interval", Rational(-5, 2), Rational(-1, 2)),
        ("ending_at_zero", S(-1), S.Zero),
    ],
)
def test_uniform_quantile_at_one_returns_upper_endpoint(name, a, b):
    U = Uniform("U_" + name, a, b)
    assert quantile(U)(S.One) == FiniteSet(b)
```